Ming Shen

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RESEARCH INTEREST

I am broadly interested in various topics related to natural language processing. My focus is on utilizing synthetic artifacts to build effective NLP systems.

EDUCATION

Ph.D. in Computer Science 2020 — 2026
Arizona State University, Tempe, Arizona
- Co-supervised by Prof. Chitta Baral and Prof. Ben Zhou.

M.S. in Computer Science 2018 — 2020
University of Southern California, Los Angeles, California
- Supervised by Prof. Xiang Ren.

B.S. in Computer Science 2014 — 2018
Worcester Polytechnic Institute, Worcester, Massachusetts

PUBLICATION

1. Relational Priors as Convergence Pressure in LLM-Based Multi-Agent Systems
Ming Shen, *Chao Shang, Sadat Shahriar, Devang Kulshreshtha, Yi Zhang, Sandesh Swamy, Yanjun Qi*
arXiv, 2026
2. BOW: Training Language Models to Reason Over Plausible Next Words
Ming Shen*, *Zhikun Xu**, *Jacob Dineen, Xiao Ye, Ben Zhou*
in Proceedings of EMNLP 2026
3. Optimizing LLM-Based Multi-Agent System with Textual Feedback: A Case Study on Software Development
Ming Shen, *Raphael Shu, Anurag Pratik, James Gung, Yubin Ge, Monica Sunkara, Yi Zhang*
COLM 2025 Workshop on AI Agents: Capabilities and Safety
4. ToW: Thoughts of Words Improve Reasoning in Large Language Models
*Zhikun Xu**, ***Ming Shen****, *Jacob Dineen, Zhaonan Li, Xiao Ye, Shijie Lu, Aswin RRV, Chitta Baral, Ben Zhou*
in Proceedings of NAACL 2025
5. QA-LIGN: Aligning LLMs through Constitutionally Decomposed QA
Jacob Dineen, Aswin RRV, Qin Liu, Zhikun Xu, Xiao Ye, ***Ming Shen****, Zhaonan Li, Shijie Lu, Chitta Baral, Muhao Chen, Ben Zhou*
in Findings of EMNLP 2025
6. Rethinking Data Selection for Supervised Fine-Tuning
Ming Shen
arXiv, 2024
7. Towards LogiGLUE: A Brief Survey and A Benchmark for Analyzing Logical Reasoning Capabilities of Language Models
Man Luo, Shrinidhi Kumbhar, ***Ming Shen****, Mihir Parmar, Neeraj Varshney, Chitta Baral, Pratyay Banerjee, Somak Aditya*
arXiv, 2023
8. Simple Yet Effective Synthetic Dataset Construction for Unsupervised Opinion Summarization

Ming Shen, *Jie Ma, Shuai Wang, Yogarshi Vyas, Kalpit Dixit, Miguel Ballesteros, Yassine Benajiba*
in Findings of EACL 2023

9. Unsupervised Pronoun Resolution via Masked Noun-Phrase Prediction
Ming Shen*, Pratyay Banerjee*, Chitta Baral*
in Proceedings of ACL-IJCNLP 2021

10. CommonGen: A Constrained Text Generation Challenge for Generative Commonsense Reasoning
Bill Yuchen Lin, Wangchunshu Zhou, ***Ming Shen****, Pei Zhou, Chandra Bhagavatula, Yejin Choi, Xiang Ren*
in Findings of EMNLP 2020

11. TriggerNER: Learning with Entity Triggers as Explanations for Named Entity Recognition
Bill Yuchen Lin, Dong-Ho Lee*,* ***Ming Shen****, Ryan Moreno, Xiao Huang, Prashant Shiralkar, Xiang Ren*
in Proceedings of ACL 2020

12. Evaluating Medical LLMs by Levels of Autonomy: A Survey Moving from Benchmarks to Applications
Xiao Ye, Jacob Dineen, Zhaonan Li, Zhikun Xu, Weiyu Chen, Shijie Lu, Yuxi Huang, ***Ming Shen****, Phu Tran, Ji-Eun Irene Yum, Muhammad Ali Khan, Muhammad Umar Afzal, Irbaz Bin Riaz, Ben Zhou*
arXiv, 2025

13. Methods and Mechanisms for Interactive Novelty Handling in Adversarial Environments
Tung Thai, ***Ming Shen****, Mayank Garg, Ayush Kalani, Nakul Vaidya, Utkarsh Soni, Mudit Verma, Sriram Gopalakrishnan, Chitta Baral, Subbarao Kambhampati, Jivko Sinapov, Matthias Scheutz*
in Proceedings of AAMAS 2023

14. An Architecture for Novelty Handling in a Multi-Agent Stochastic Environment: Case Study in Open-World Monopoly
Tung Thai, ***Ming Shen****, Neeraj Varshney, Sriram Gopalakrishnan, Utkarsh Soni, Matthias Scheutz, Chitta Baral, Jivko Sinapov*
in Proceedings of AAAI 2022 Symposium: Designing Artificial Intelligence for Open Worlds

EXPERIENCE

Applied Scientist — Sep. 2026 — Now
Amazon Bedrock, Seattle, Washington

Applied Scientist Internship — Sep. 2025 — Dec. 2025
Amazon Bedrock, Seattle, Washington

Applied Scientist Internship — May 2024 — Aug. 2024
Amazon Bedrock, New York, New York

Applied Science Internship — May 2023 — Aug. 2023
Amazon Bedrock, New York, New York

Applied Scientist Internship — May 2022 — Aug. 2022
Amazon Comprehend, New York, New York

Graduate Teaching Associate Assistantship — Aug. 2022 — May 2025
School of Computing and Augmented Intelligence, ASU, Tempe, Arizona

Graduate Research Associate Assistantship Aug. 2020 — May 2022
School of Computing and Augmented Intelligence, ASU, Tempe, Arizona

Graduate Student Worker Nov. 2019 — May 2020
Information Science Institute, USC, Los Angeles, California

PROJECT

Demo project for *ACL 2019 System Demonstrations* paper: AlpacaTag: Active Learning-based Crowd Annotation Framework for Sequence Tagging

AWARD HONOR

University Doctoral Fellowship, Arizona State University 2020 — 2026
CIDSE Doctoral Fellowship, Arizona State University 2020

SERVICE

Reviewer: ACL Rolling Review, ACL, EMNLP, COLING